Overlap Guided Adaptive Fractionation - Clincial Trial Version

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Abstract

Ipsum thingy

1 Introduction

In our former work we established an adaptive fractionation algorithm that computes the optimal doses for each fraction when the PTV overlaps with an OAR. A

2 Methods and Materials

2.1 Patient selection and treatment plans

The data for this analysis was yielded from all patients with abdominal primary or secondary tumours, who received SBRT with MRgRT at the MR-linac (MRIdian, ViewRay) in our department between 2019 and 2024. Patients were included in the analysis, if the PTV was located in direct proximity to the dose-limiting OARs bowel, stomach, and/or duodenum, overlapping with at least one of these organs in at least one fraction of treatment, and thus necessitating a PTV coverage compromise to allow for OAR sparing. Furthermore, to qualify for inclusion, SBRT had to have been administered in 5 fractions with a minimum dose of 6 Gy per fraction, regardless of the prescribed isodose level. Treatment courses were excluded, if more than one PTV was irradiated in the same plan.

Potential improvement: multi-target?

We isolated a total of 58 treatment courses that met these criteria and extracted volumetric and dosimetric information on GTV, PTV and relevant OARs for a total of 6 treatment plans per course (the initial plan based on the planning MRI, plus the 5 delivered plans) from our TPS. Treatment planning and delivery for all included patients followed institutional protocols: Each patient underwent either MR-only, or combined MR- and CT-simulation, as well as daily MR imaging for online adaptive radiotherapy. Tumours and OARs within a 2 cm margin around the GTV were recontoured prior to each fraction according to institutional guidelines. Adaptive treatment plans were generated daily to account for inter-fractional anatomical changes, with prescription doses remaining constant across all fractions.

How?

What's the GTV-PTV margin for MR-Linac SBRT?

Overlap between PTV and the relevant, dose-limiting OARs was calculated by creating an overlap structure for each overlapping OAR with the PTV and measuring its volume. If more than one dose-limiting OAR showed overlap, all separate overlap volumes were added together to create a total overlap volume. All considerations on adaptive fractionation below were made based on this total overlap volume.

So it's been used already???

Additionally, 6 of the 58 SBRT courses underwent retrospective re-planning based on the adapted doses calculated by way of the algorithm detailed in the following sections.

2.2 Quantification of Geometric and Dosimetric Variability

To characterize inter-fractional anatomical variability, we quantified the volumetric overlap between the planning target volume (PTV) and the organs at risk (OARs). Let o_t denote the overlapping PTV volume with the relevant OAR on treatment day t, measured in cubic centimeters (cc).
As an approximate measure of the potential loss in PTV coverage during fraction t, we defined a cost function c_t^{app} as

$$c_t^{\text{app}}(o_t) = o_t \cdot (d_t - d_{\min}) \tag{1}$$

Here, d_t is the prescribed dose to the PTV in fraction t, and $d_{\min}$ represents the maximum deliverable dose to the overlapping region without violating per-fraction OAR constraints. The cost c_t^{app} is therefore proportional to the excess of the prescribed dose above the deliverable dose $d_{\min}$ within the overlap region.

Figure 1 schematically illustrates this concept using a dose–volume histogram (DVH) of the PTV for a prescription dose $d_t > d_{\min}$. The area above the DVH curve represents the true underdosage of the PTV due to OAR dose limitations. The red rectangle corresponds to the approximate cost area defined by $c_t^{\text{app}}(o_t)$. The cost is expressed in units of ccGy, meaning that an improvement of 1 ccGy corresponds to 1 cc of the PTV receiving 1 Gy more dose.
In addition, two triangular contributions improve the approximation of the area above the curve. The first, located left of $d_{\min}$, corresponds to a baseline underdosage that is always present even when $d_{\min}$ is prescribed. The second, to the right of $d_{\min}$, represents additional underdosage caused by finite dose gradients at the PTV–OAR interface when delivering doses above $d_{\min}$. The latter is quantified as

$$c_t^{tri} = \frac{m_t}{2} \cdot (d_t - d_{\min})^2 \tag{2}$$

where m_t is a slope parameter describing the steepness of the PTV DVH between $d_{\min}$ and d_t, measured in cc per Gy.

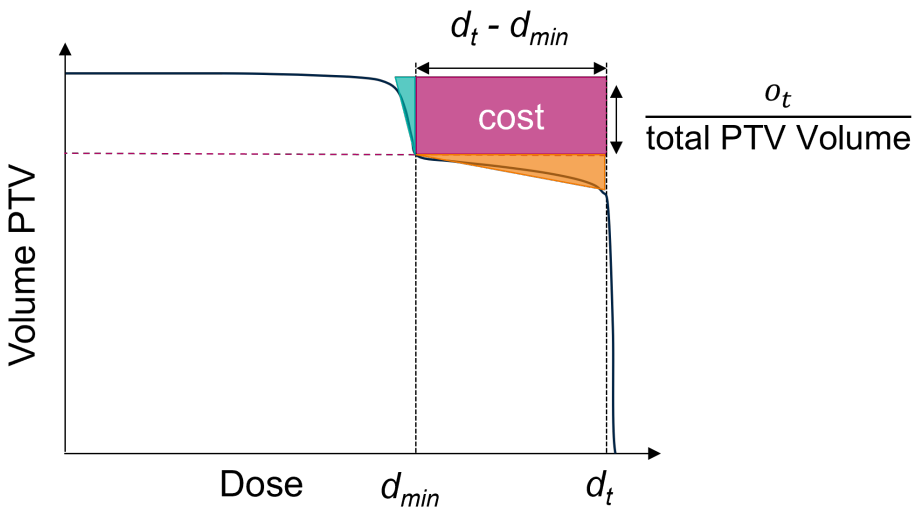
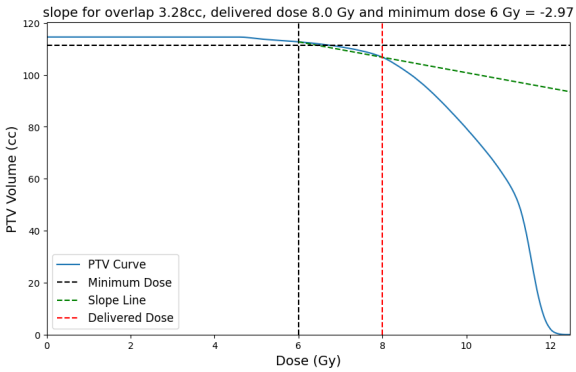


Figure 1: Schematic illustration of PTV underdosage estimation. The blue line represents the PTV DVH. The dashed red line indicates the overlap volume. The red box corresponds to the cost $c_t^{\text{box}}(o_t)$, while the green triangle represents a PTV underdosage that is always present, and the orange area represents an additional underdosage introduced by dose gradients for prescription doses larger than $d_{\min}$.

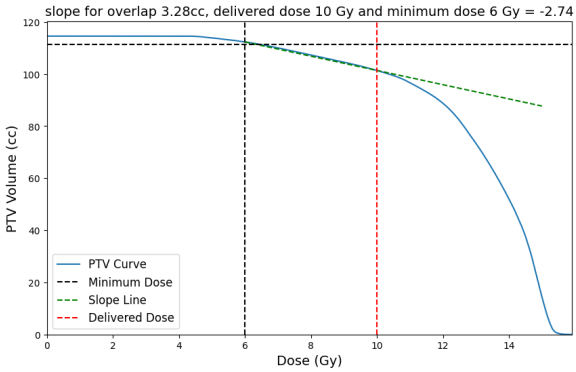
To estimate the PTV underdosage for a prescribed dose d_t, we introduced a cost function that combines the two underdosage functions d_t^{box} and d_t^{tri}. The total underdosage cost is expressed as:

$$c_t(o_t, m_t) = o_t \cdot (d_t - d_{\min}) + \frac{m_t}{2} \cdot (d_t - d_{\min})^2 \tag{3}$$

Empirical analysis of clinical treatment plans showed that the combined representation by c_t^{box} and c_t^{tri} provides an accurate approximation of the PTV underdosage that arises when doses larger than $d_{\min}$ are prescribed.



(a) DVH for a moderate dose of 8 Gy. Vertical lines represent the minimum deliverable dose (black) and the prescribed dose (red). The horizontal black line indicates the overlap volume, and the tilted green line approximates the dose gradient between these two doses.



(b) DVH for a high dose of 10 Gy. Steepness between $d_{\min}$ and d_t is comparable to that in panel (a), demonstrating that for a given anatomic configuration, the DVH slope m_t depends primarily on the overlap o_t rather than on the prescribed dose.

In Figures 2a and 2b two DVHs for the same fraction are shown. For both plans, the steepness between d_{min} and d_t is similar, even though we deliver two different doses, showing that the steepness m_t is mostly dependent on the overlap o_t and not the dose.

Whereas the overlap volume o_t can be estimated based on the MRI scan of fraction t once the contours have been adapted, the slope m_t can only be evaluated once a dose distribution is computed. For a practical adaptive fractionation method, m_t must be estimated a priori from the overlap geometry before treatment planning.

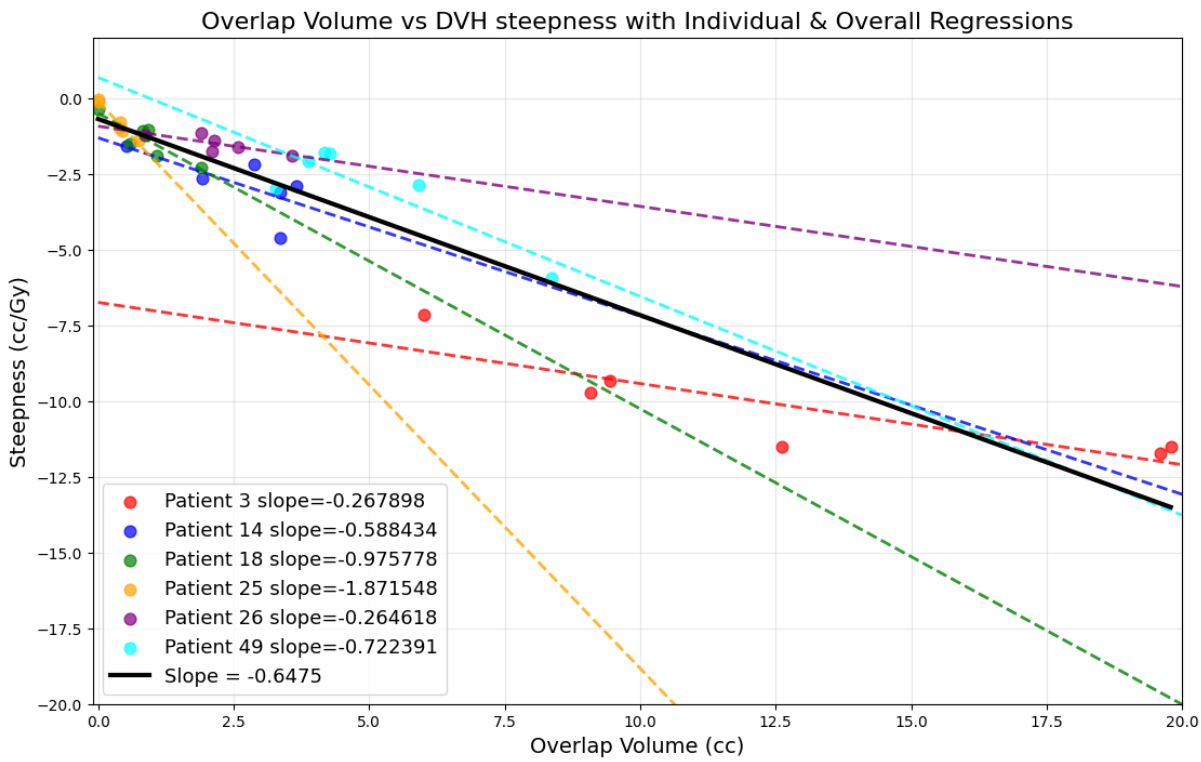


Figure 3: Correlation between overlap volume o_t and DVH steepness m_t for six patients. Each patient contributed six data points (one planning scan and five treatment fractions). Individual patient regressions are shown as colored lines, and the overall regression across all patients is shown in black. A negative slope indicates that larger overlap volumes are associated with steeper DVH fall-offs.

To enable estimation of m_t before treatment planning, the empirical relationship between overlap volume o_t and DVH steepness m_t was analyzed for six patients (Figure 3). Each patient contributed six data

points—one planning scan and five treatment fractions—yielding a total of 36 measurements. A roughly linear correlation was observed, indicating that anatomical overlap is a reliable predictor of the steepness of the PTV DVH.

Assuming a linear dependence, the slope parameter m_t is expressed as:

$$m(o_t) = a - b \cdot o_t, \tag{4}$$

where a and b denote the intercept and slope of the regression, respectively. The estimated parameters were approximately $a = -0.65$ and $b = -0.65$.
For practical implementation, and because the cost must vanish when the prescribed dose equals $d_{\min}$, the intercept a is set to zero. The resulting simplified expression is

$$m(o_t) = -b \cdot o_t. \tag{5}$$

Substituting the absolute value of Eq. (4) into Eq. (3) yields a fully predictive cost function that depends only on the overlap volume:

$$c_t(o_t) = o_t \cdot (d_t - d_{\min}) + \frac{m(o_t)}{2}(d_t - d_{\min})^2 \tag{6}$$

This formulation allows the dosimetric impact of inter-fractional geometric variations to be approximated directly from anatomical overlap information, without requiring individualized treatment planning for each possible configuration.

Aggregated Cost and Illustrative Example

The overall quality of an adaptive fractionation (AF) strategy can be evaluated by aggregating the daily costs over the full treatment course. The total accumulated cost is defined as:

$$c_{total} = \sum_{t=1}^{F} c_t(o_t) = o_t \cdot (d_t - d_{\min}) + \frac{m(o_t)}{2}(d_t - d_{\min})^2 \tag{7}$$

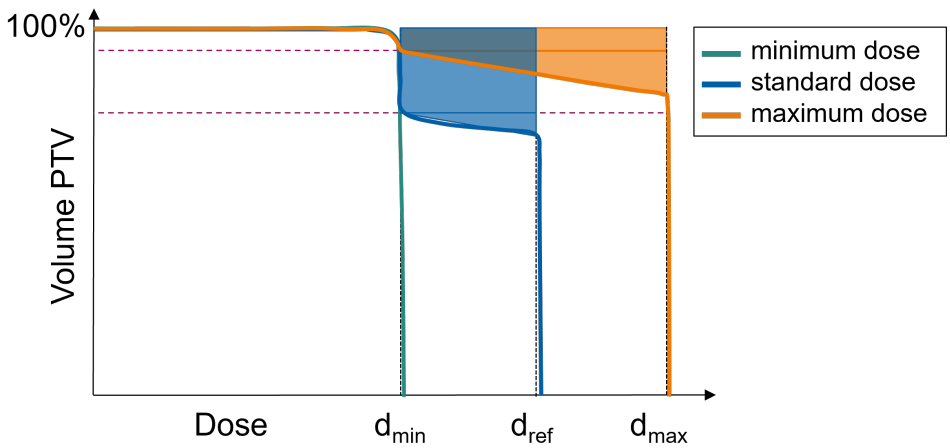


Figure 4: Comparison of DVHs for different dose prescriptions and overlap scenarios. The green line shows the minimum deliverable dose $d_{\min}$ that ensures no underdosage regardless of overlap. The blue line represents a uniform dose d_{ref} delivered under high-overlap conditions, resulting in significant underdosage (blue box). The orange line represents a high dose delivered on a low-overlap day, yielding moderate underdosage despite the higher dose (orange box). Dashed red lines indicate overlap volumes.

To illustrate the rationale for AF based on overlap further, Figure 4 illustrates two fractions with different overlap volumes.
Suppose the overlap volume is $o_1 = 10\,\text{cc}$ in the first fraction and $o_2 = 3\,\text{cc}$ in the second fraction. A uniform fractionation scheme prescribes a constant dose of $d_{\text{ref}} = 8$ Gy for each fraction, with an OAR constraint of $d_{\min} = 6$ Gy.

The total cost for uniform fractionation is then:

Uniform Fractionation $d_1 = d_2 = 8$ Gy:

$$c_{\text{total}} = 10 \text{ cc} \cdot (8-6) \text{ Gy} + \frac{0.65 \cdot 10 \text{ cc}}{2}(8-6)^2 \text{ cc} + 3 \text{ cc} \cdot (8-6) \text{ Gy} \frac{0.65 \cdot 3 \text{ cc}}{2}(8-6)^2 \text{ Gy} = \mathbf{42.9 \text{ ccGy}}. \quad (8)$$

In contrast, an adaptive strategy may deliver 6 Gy in the first (unfavorable) fraction and 10 Gy in the second (favorable) fraction. Since the 6 Gy dose fully respects OAR limits, no underdosage occurs in fraction one, resulting in zero cost. For the second fraction, the small overlap of 3 cc allows delivery of 10 Gy. The total cost is thus:

Adaptive Fractionation: $d_1 = 6$ Gy, $d_2 = 10$ Gy,

$$c_{\text{total}} = 0 \text{ ccGy} + 3 \text{ cc} \cdot (10-6) \text{ Gy} + \frac{0.65 \cdot 3 \text{ cc}}{2}(10-6)^2 \text{ Gy} = \mathbf{27.6 \text{ ccGy}}. \quad (9)$$

In this simplified example, adaptive fractionation reduces the cumulative cost by approximately **35%** compared with uniform fractionation. Even though the underdosage in the high-dose fraction slightly increases, the overall performance improves because the large underdosage on the high-overlap day is entirely avoided by delivering the minimum dose d_{min}.

Probability distribution for overlap volumes

The central challenge in AF is that future anatomical configurations are not known in advance. Adaptive fractionation relies on anticipated overlap volumes for future fractions for determining the optimal dose in the current fraction. In this work, we assume that the daily overlaps follow a normal distribution $o_t \sim \mathcal{N}(\mu, \sigma^2)$, which is updated iteratively during treatment. As in [1], we estimate the patient-specific parameters μ and σ based on observed overlap values up to the current fraction.
Because mean overlap volumes can differ significantly across patients, we use the maximum likelihood estimator to compute the patient-specific mean, updating it with each observed value:

$$\mu_t = \frac{1}{t+1} \sum_{\tau=0}^{t} o_\tau \quad (10)$$

Here, o_0 denotes the overlap volume observed in the planning MR.

The standard deviation estimated from the observed overlap volumes up to fraction t is given by

$$\sigma_t^{pat} = \sqrt{\frac{1}{t+1} \sum_{\tau=0}^{t} (o_\tau - \mu_t)^2} \quad (11)$$

Since the standard deviation strongly influences the adaptive fractionation policy, we adopt a Bayesian approach to regularize early estimates. Specifically, we use a population-based gamma prior for σ:

I still think this is too broad

$$f(\sigma; k, \theta) = \frac{1}{\Gamma(k)\theta^k} \sigma^{k-1} \exp\left(\frac{-\sigma}{\theta}\right) \quad (12)$$

The hyperparameters k (shape) and θ (scale) were estimated empirically from the distribution of standard deviations across our retrospective cohort of 58 patients, yielding $k = 1.07$ and $\theta = 0.78$.

Given this prior and the observed overlaps up to fraction t, we compute the maximum a posteriori estimate of σ_t. The likelihood is based on a normal model with unknown variance, and the resulting maximum a posteriori estimate can be obtained numerically by maximizing the posterior:

$$\sigma_t = \underset{\sigma}{\text{argmax}} \left[\frac{\sigma^{k-1}}{\sigma^{t-1}} \exp(-\theta\sigma) \exp\left(\frac{(\sigma_t^{pat})^2}{2\frac{\sigma^2}{t}} \right) \right] \quad (13)$$

The a posteriori estimator balances patient-specific observations with population-informed regularization, yielding more robust estimates, especially in the early treatment phase when few overlap volumes were observed.

Markov Decision Process Model

To solve the adaptive fractionation problem under uncertainty about future overlaps, we model the decision process as a Markov Decision Process (MDP) similarly to [1]. The goal is to find a dose selection policy that minimizes the expected cumulative cost over F treatment fractions, while delivering the prescribed cumulative dose to the PTV.

The MDP is described by states, actions, state transitions and cost, which are given by:

State.
In each fraction t, the state is defined as a tuple $s_t = (o_t, D_t)$ where:

- o_t is the observed PTV–OAR overlap volume in fraction t,
- $D_t = \sum_{\tau}^{t-1} d_\tau$ is the cumulative dose delivered to the PTV up to (but not including) fraction t.

This state encapsulates the current geometric condition and treatment history, both of which influence the optimal dose decision.

Action.
The action is the dose d_t prescribed to the PTV in fraction t. The action space is bounded:

$$d_t \in [d_{\min}, d_{\max}] \tag{14}$$

In this work the lower bound d_{min} is set to 6 Gy, equivalent to the dose constraints in the OAR. The upper bound depends on the prescription dose as stated in the protocol.

State Transition.
After applying dose d_t in state $s_t = (o_t, D_t)$, the state transitions to a new state $s_{t+1} = (o_{t+1}, D_{t+1})$, where:

- $o_{t+1} \sim \mathcal{N}(\mu_t, \sigma_t^2)$ is drawn from the patient-specific normal distribution,
- $D_{t+1} = D_t + d_t$ is the updated cumulative PTV dose.

Since o_{t+1} is stochastic, the transition is probabilistic.

Cost.
The immediate cost in each fraction is defined by the formerly introduced cost:

$$c_t(o_t) = o_t \cdot (d_t - d_{\min}) + \frac{m(o_t)}{2}(d_t - d_{\min})^2 \tag{15}$$

In addition, we impose a terminal cost c_{F+1} to ensure that the total prescribed PTV dose D_{pres} is delivered exactly:

$$c_{F+1} = \begin{cases} 0 & \text{if } D_F + d_F \geq D_{\text{pres}} \\ \infty & \text{otherwise} \end{cases} \tag{16}$$

Dynamic Programming Algorithm.

The optimal policy $\pi_t(o, D)$ assigns to each state (o, D) the optimal action that minimizes the expected cumulative cost until the end of treatment in fraction F. To compute the optimal policy for adaptive fractionation, we apply dynamic programming (DP), which recursively computes the optimal dose to deliver in each fraction by minimizing the sum of immediate cost and expected future cost [2]. DP is based on the value function $v_t(o, D)$, which quantifies the expected future cost when starting from state (o, D) in fraction t and following the optimal policy from there on. Optimal policy and value function can then be calculated in a backward recursion starting in the last fraction $t = F$ and iterating backwards to $t = 1$. One iteration of DP is given by

$$v_t(o, D) = \min_{d \in [d_{\min}, d_{\max}]} \left[c_t(o) + \sum_{o'} P(o') v_{t+1}(o', D + d) \right] \tag{17}$$

$$\pi_t(o, D) = \underset{d \in [d_{\min}, d_{\max}]}{\operatorname{argmin}} \left[c_t(o) + \sum_{o'} P(o') v_{t+1}\left(o', D + d\right) \right] \tag{18}$$

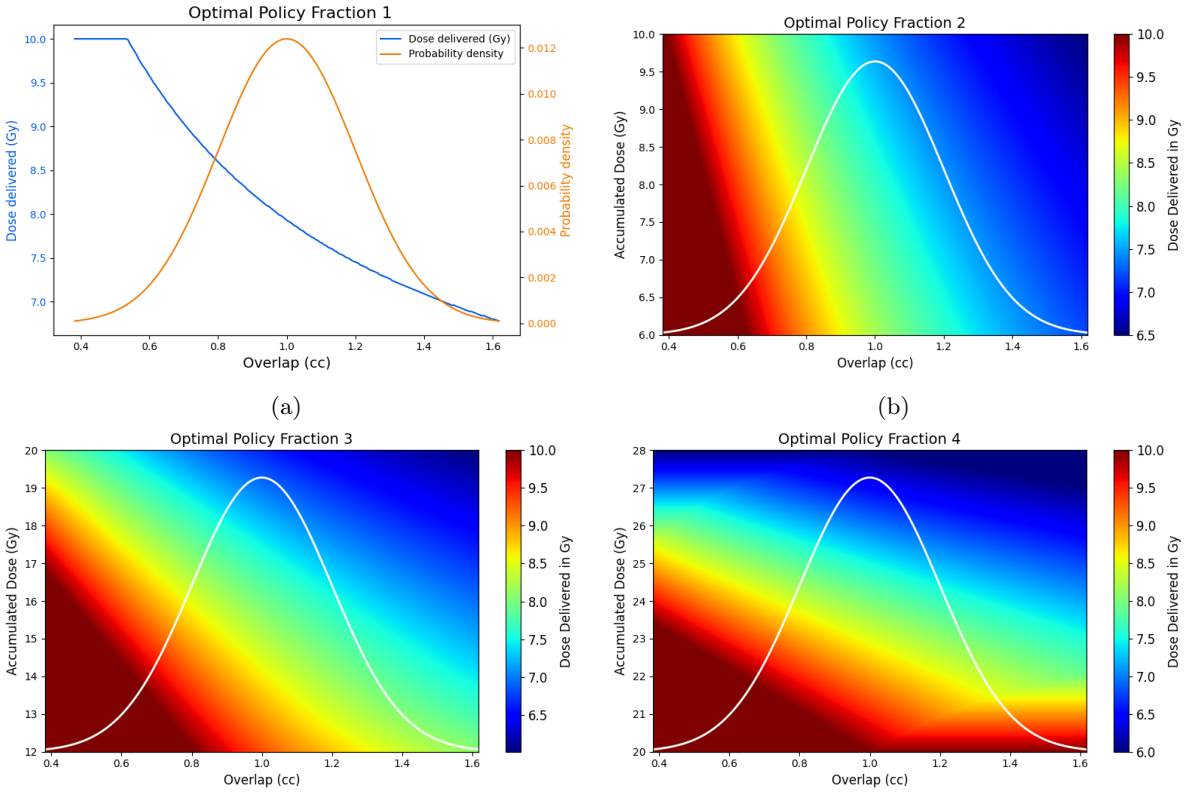
where the value function is initialized through the terminal cost, i.e. $v_{F+1}(o, D) = c_{F+1}(o, D)$. Here, $P(o')$ denotes the probability distribution over the unknown overlap volume in the next fraction, which is given by the normal distribution $\mathcal{N}(\mu, \sigma^2)$. The sum over o' accounts for the stochastic nature of future anatomy.

To compute v_t and π_t in practice, we discretize both the state space, i.e. D and o, and the action space $[d_{min}, d_{max}]$. Note that we make the approximation that the estimates of μ and σ are not part of the state of the MDP but are assumed to be constant over future fractions. To apply AF to a given patient with a given sequence of overlap volumes, we update μ and σ in each fraction as described in section 2.2 and recompute the optimal policy.

3 Optimal Policy Discussion

We evaluated adaptive fractionation on retrospective data of 58 patients. The analysis resulted in three patient types which were thoroughly discussed to assess the proposed doses that will be used as look-up tables during the clinical trial. Here we present the evaluation of these patient types, i.e. patients with **small** overlap volumes and variation, patients with **average** overlap volume and variation, and patients with **large** overlap volumes and variation. For all patients we present the optimal policy, i.e. the dose to deliver for all possible cases for a 5 fraction treatment with a 40 Gy prescription, where the minimum dose to deliver each fraction is 6 Gy (the dose constraint in the OAR) and the maximu dose is 10 Gy. Only fractions 1-4 are shown as in the final 5th fraction we always deliver the residual dose to reach the 40 Gy prescription dose, independent of the overlap observed.

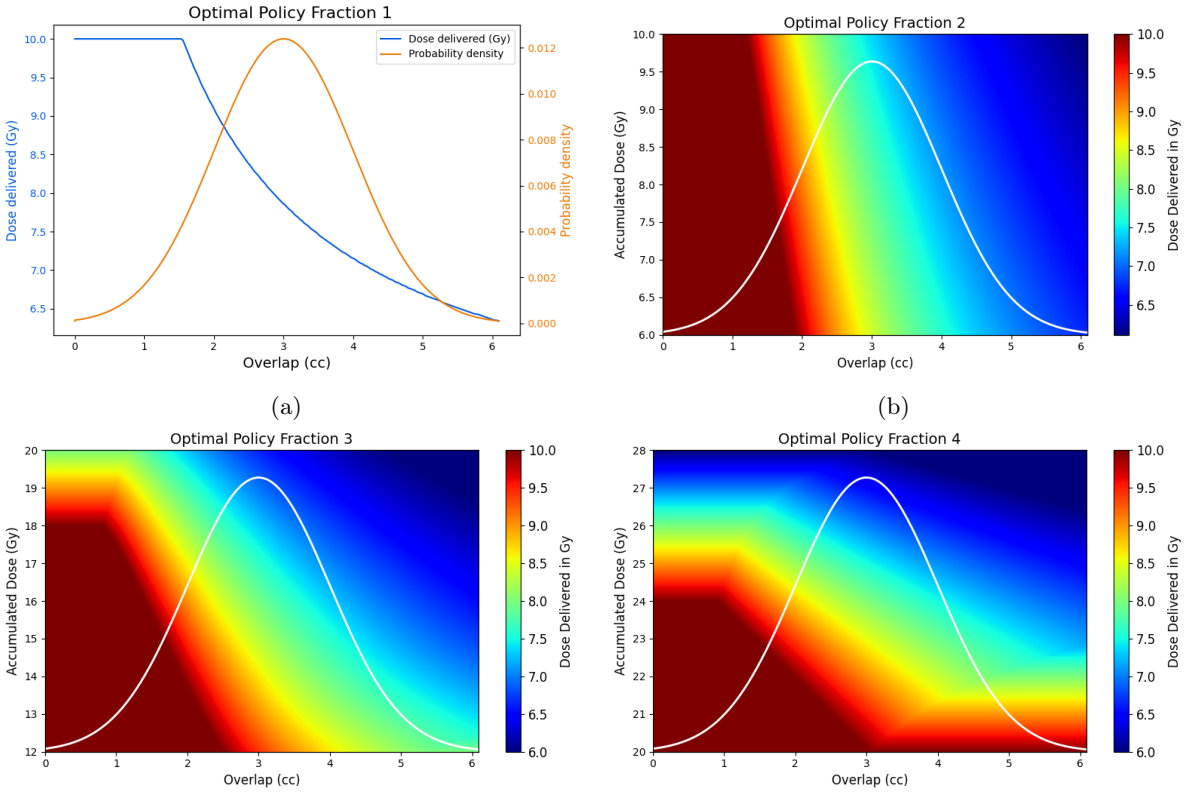
Small Volumes and Variation Patients Patients with small overlap volumes and variation presented with a mean overlap of 1 cc and a std of 0.2 cc. The resulting dose proposals are shown in Figure 5



(a) (b) (c) (d)

Figure 5: Optimal policies and values for a patient with small overlap and standard deviation. The plots show the optimal policy for fractions one through four. For the first fraction the probability distribution over the overlaps is shown in orange. For the remaining fractions it's shown in white.

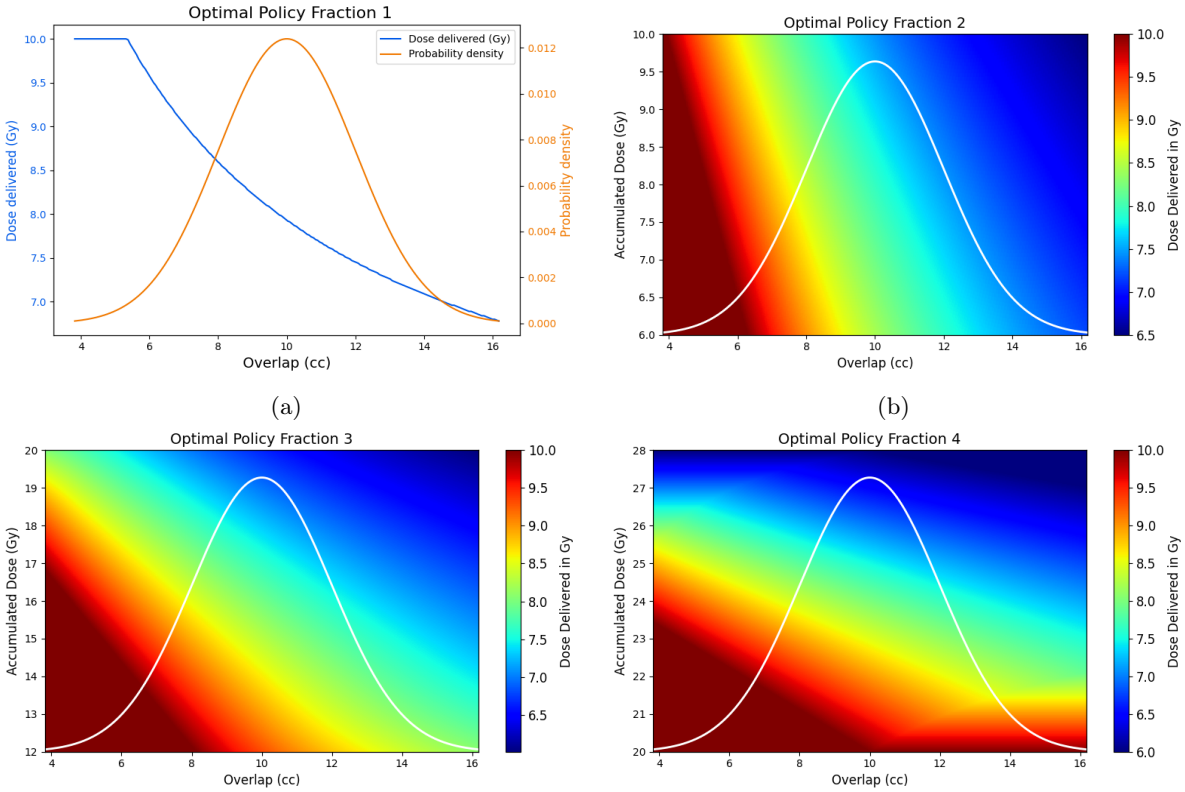
Average Volumes and Variation Patients Average patients presented with a mean overlap of 3 cc and a std of 1 cc. The resulting dose proposals are shown in Figure 6



(a) (b) (c) (d)

Figure 6: Optimal policies and values for a patient with average overlap and standard deviation. The plots show the optimal policy for fractions one through four. For the first fraction the probability distribution over the overlaps is shown in orange. For the remaining fractions it's shown in white.

Large Volume and Variation Patients Patients with large overlap volumes and variation presented with mean overlap of 10 cc and a std of 2 cc. The resulting dose proposals are shown in Figure 7



(a) (b) (c) (d)

Figure 7: Optimal policies and values for a patient with large overlap and standard deviation. The plots show the optimal policy for fractions one through four. For the first fraction the probability distribution over the overlaps is shown in orange. For the remaining fractions it's shown in white.

4 Selected Results

First we would like to assess, what benefit distribution we obtain under AF using different dose steps in the action space. For that we retrospectively analyzed all 58 patients in the GitHub repository. Since not all patients have the same prescription dose we adapted the maximum dose d_{max} according to the prescribed dose. If the patients have a prescription dose between ≤ 40 Gy, the maximum dose per fraction is set to 10 Gy. If the prescription dose is between 40 and 45 Gy, the maximum dose is set to 11 Gy. If the prescription dose is larger than 45 Gy, the maximum dose is set to 12 Gy.

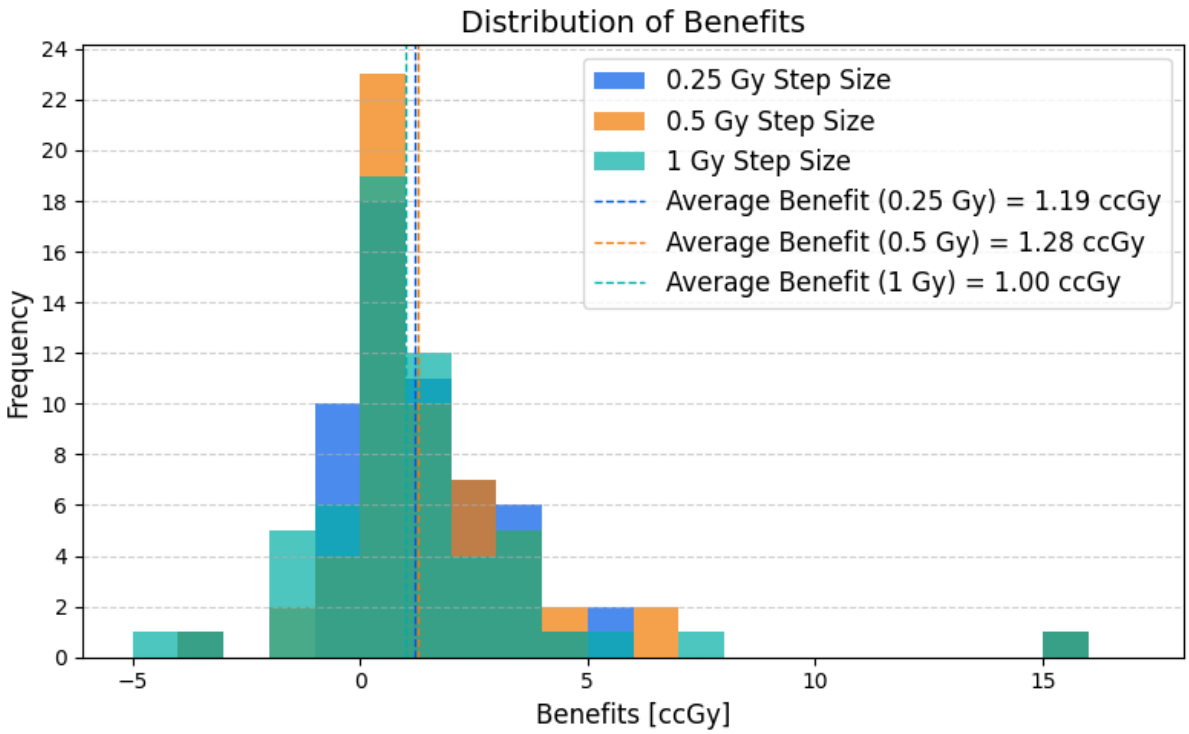


Figure 8

As we can see, the finer the step size, the larger the average benefit. With smaller step size, the probability mass shifts to the left. However, the 0.5 Gy steps show a slightly better benefit than the 0.25 Gy steps.

To assess the benefits achieved by each patient we plot a comparison between the new AF treatment and an optimal upper bound that can be reached when all overlaps are known a priori for 0.5 Gy dose steps.

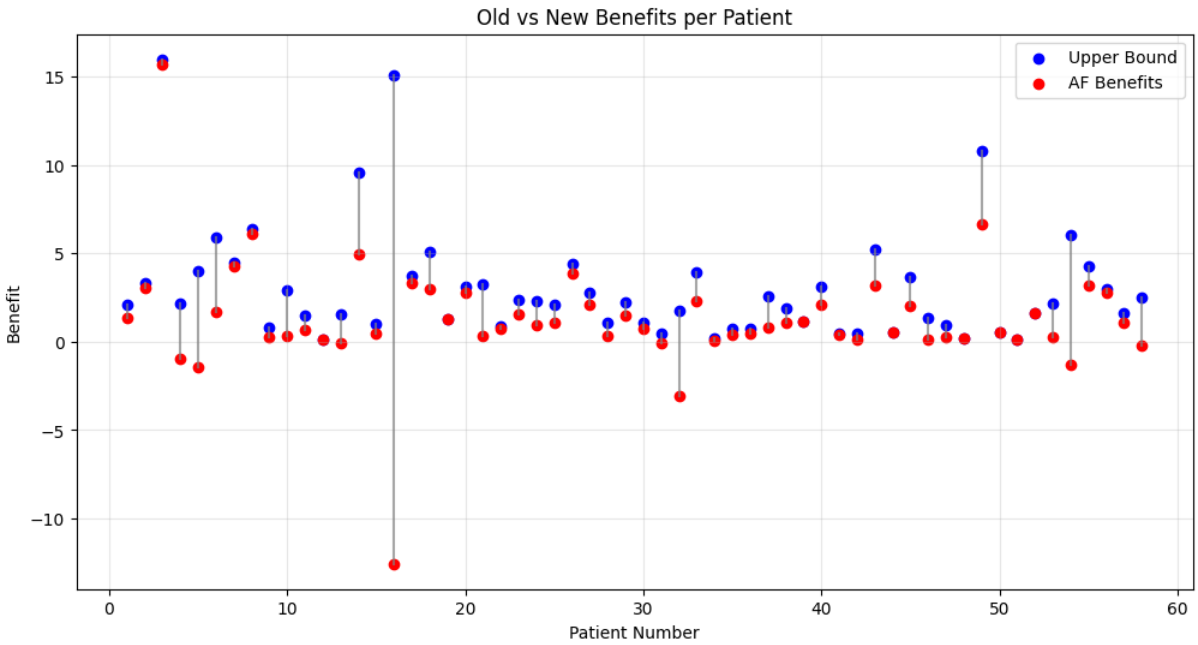


Figure 9: 0.5 Gy steps

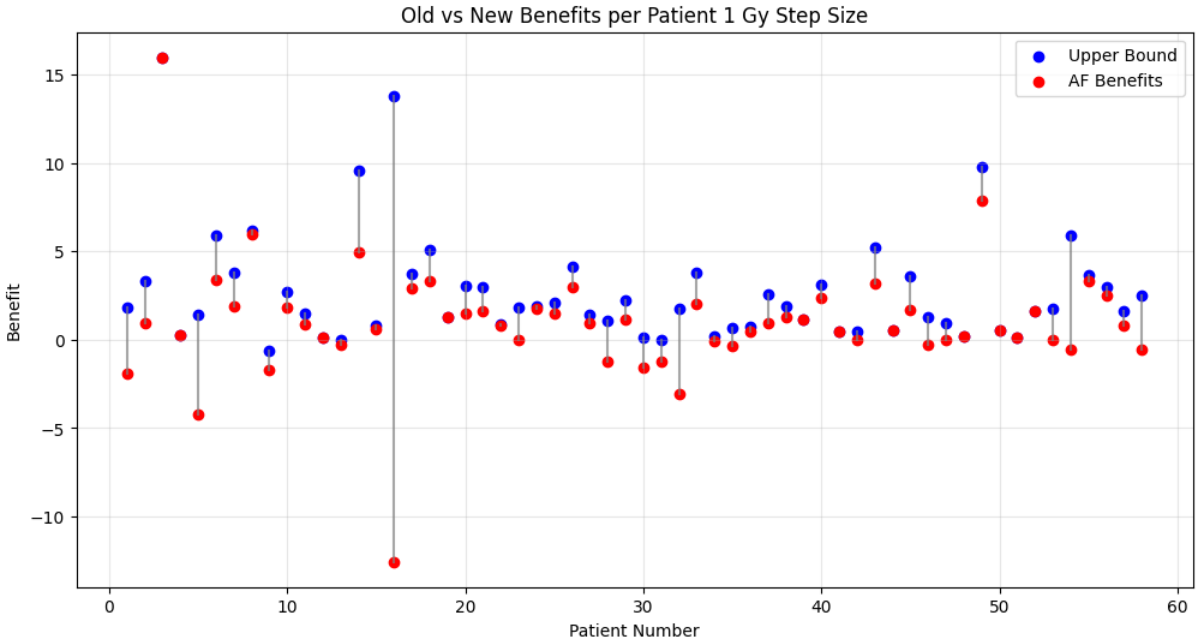


Figure 10: 1 Gy steps

5 Appendix

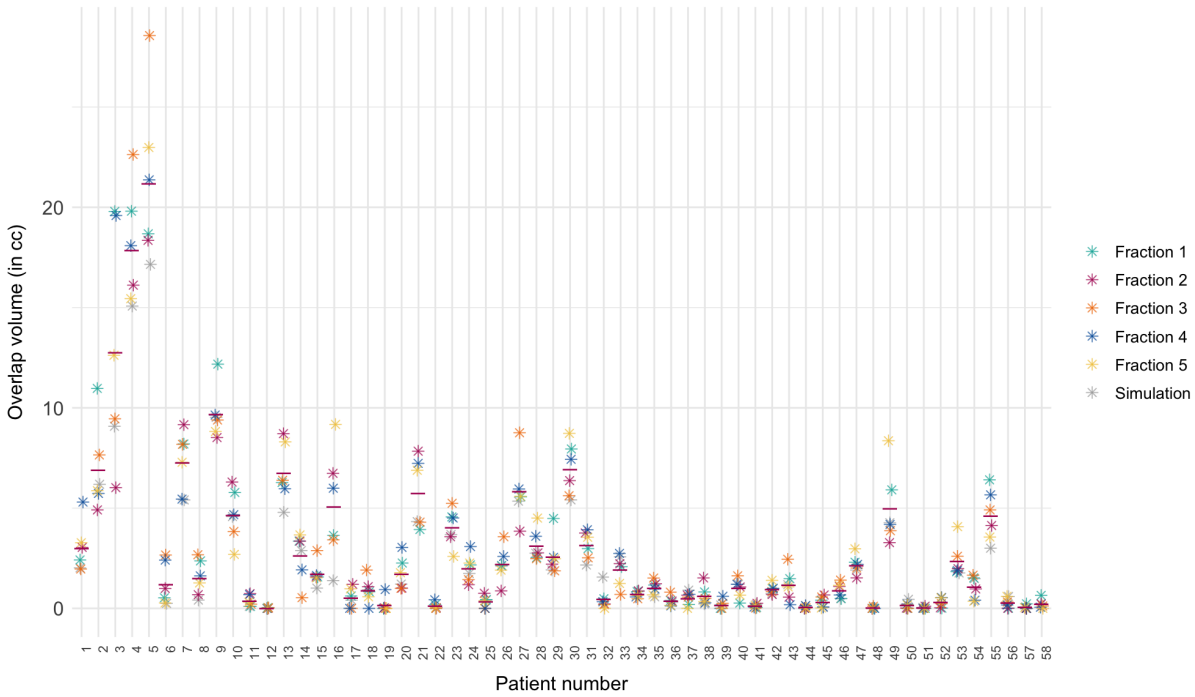


Figure 11: Variation in overlap volumes for all patients. Each star represents the total calculated overlap volume for one fraction, including the initial simulation. The red line represents the mean overlap volume throughout each treatment course.

References

[1] Pérez Haas, Y., Ludwig, R., Dal Bello, R., Tanadini-Lang, S. & Unkelbach, J. Adaptive fractionation at the mr-linac. *Physics in Medicine & Biology* **68**, 035003 (2023).

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[2] Sutton, R. S. & Barto, A. G. *Reinforcement Learning: An Introduction* (A Bradford Book, Cambridge, MA, USA, 2018).